

Field and Service Robotics — Final Technical Report

**Modeling and Multi-Strategy Control of a
Variable-Length Wheeled Inverted Pendulum Robot**

RoTino: a Wheeled Biped Balancing Platform

Aldo Rosicarelli

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1 Introduction

Wheeled biped robots occupy a particularly interesting niche in the landscape of field and service robotics: they combine the high mobility and energy efficiency of wheeled platforms with the postural versatility of legged systems. By adjusting their centre of mass (CoM) height through variable leg length, they can negotiate uneven terrain, step over obstacles, or adopt a compact stance for confined environments. The archetype is the family of *Variable-Length Wheeled Inverted Pendulum* (VL-WIP) robots — two-wheeled self-balancing machines whose legs can change length to modulate the height at which the upper-body mass is carried.

The control problem is inherently challenging because the system is *underactuated*: the pitch angle of the torso is not directly actuated, but must be controlled indirectly through the centroidal coupling with the wheel acceleration. This is analogous to the classical inverted-pendulum-on-a-cart problem, but with the additional complexity of a multi-link leg mechanism, configuration-dependent inertia, and the variable pendulum length that makes the dynamics time-varying even at equilibrium.

The project selected for this course is **RoTino**, a simulated wheeled biped modelled after the platform described by Cui et al. [1]. The development follows a two-stage methodology:

1. **MATLAB/Simulink prototyping.** The Euler–Lagrange dynamics are derived symbolically, and three control laws — cascade PID, LQR + MPC, and super-twisting SMC — are implemented and validated in ODE45/Simulink simulations of the planar model.
2. **ROS 2 / Gazebo validation.** The controllers are ported to Python ROS 2 nodes and tested in closed-loop Gazebo simulation of a fully articulated 3D URDF robot model, including sensor noise, actuator dynamics, and contact physics.

The choice is motivated by several factors:

- **Rich nonlinear underactuated dynamics.** A four-DOF planar chain with only three actuators requires careful treatment of the actuation matrix and its singularities.
- **Cascade control hierarchy.** The natural time-scale separation between balance (~ 500 Hz) and locomotion (~ 100 Hz) makes the system an ideal testbed for hierarchical control.
- **Comparison across control paradigms.** Three qualitatively different controllers are benchmarked on the same model, enabling a principled comparison of performance, robustness, and implementation complexity.
- **Software-in-the-loop validation.** The full pipeline from symbolic derivation to closed-loop ROS 2 simulation bridges theory and practice.

The report is structured as follows. Section 2 analyses the topology and degrees of freedom. Section 3 derives the dynamic model. Sections 4, 5, and 6 describe the three control strategies. Section ?? presents the simulation and benchmarking framework. Section 7 discusses the results. Section 9 concludes.

2 Topology Analysis and Degrees of Freedom

2.1 Mechanical Architecture

The robot (Figure 1) is a planar kinematic chain comprising four rigid bodies:

1. **Wheel** ($r = 0.127$ m, $m_w = 3.5$ kg, $I_w = 0.1$ kg m²): rolling without slipping on the ground.

2. **Shank link** ($l_1 = 0.45$ m, $m_1 = 1.2$ kg): connects the wheel axle to the knee joint.
3. **Thigh link** ($l_2 = 0.45$ m, $m_2 = 5.3$ kg): connects the knee to the hip joint.
4. **Torso** ($l_3 = 0.35$ m, $m_3 = 60.0$ kg, wheel track $d = 0.63$ m): the upper body, connected to the hip.

All link inertias are modelled as slender rods: $I_k = \frac{1}{12}m_k l_k^2$.

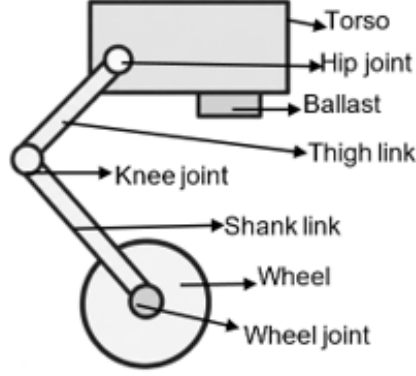


Figure 1: Schematic of the RoTino platform: torso, hip joint, ballast, thigh link, knee joint, shank link, wheel and wheel joint.

2.2 Grübler Constraint and DOF Count

A planar system of $n_b = 4$ bodies has $3n_b = 12$ unconstrained DOFs. Table 1 summarises the constraints that reduce this count to 4.

Table 1: Constraint analysis for the planar RoTino chain.

Constraint	Description	Equations	Holonomic?
Wheel-ground contact	Fixed contact height $y = r$	1	Yes
Pure rolling	$\dot{x} = r \dot{\phi}_w$	1	No (Pfaffian)
Wheel-shank revolute	2 translational DOFs locked	2	Yes
Shank-thigh revolute (knee)	2 translational DOFs locked	2	Yes
Thigh-torso revolute (hip)	2 translational DOFs locked	2	Yes
Total constraints		8	
Resulting DOF	$12 - 8$	4	

The generalised coordinates, following Cui et al. (Table 1), are:

$$\mathbf{q} = [q_w \quad q_1 \quad q_2 \quad q_3]^T \quad (1)$$

where q_w is the wheel rotation angle (encoding horizontal displacement $s = R_w q_w$ via rolling), q_1 is the absolute shank angle from the vertical, q_2 the absolute thigh angle, and q_3 the absolute torso angle.

2.3 Full 3D Model: Seven-DOF Formulation and Null-Space Projection

For the 3D Gazebo simulation, the full robot is described by $n = 7$ generalised coordinates:

$$\mathbf{q}_{3D} = [x, y, \psi, \theta, l, \phi_R, \phi_L]^T \quad (2)$$

where (x, y) is the axle midpoint on the ground plane, ψ is the heading (yaw), θ is the body pitch, l is the variable leg length, and $\phi_{R,L}$ are the right/left wheel rotations.

Three Pfaffian (non-holonomic) rolling constraints $A(\mathbf{q})\dot{\mathbf{q}} = 0$ enforce pure rolling without slipping:

$$-\dot{x} \sin \psi + \dot{y} \cos \psi = 0 \quad (\text{no lateral slip}) \quad (3)$$

$$(\dot{x} \cos \psi + \dot{y} \sin \psi) + \frac{d}{2}\dot{\psi} - r\dot{\phi}_R = 0 \quad (4)$$

$$(\dot{x} \cos \psi + \dot{y} \sin \psi) - \frac{d}{2}\dot{\psi} - r\dot{\phi}_L = 0 \quad (5)$$

These reduce the system to $m = 4$ independent pseudo-velocities $\mathbf{u} = [\dot{\phi}_R, \dot{\phi}_L, \dot{\theta}, \dot{l}]^T$, related to $\dot{\mathbf{q}}$ via the null-space basis $S(\mathbf{q}) \in \mathbb{R}^{7 \times 4}$ such that $\dot{\mathbf{q}} = S(\mathbf{q})\mathbf{u}$:

$$S(\mathbf{q}) = \begin{bmatrix} \frac{r}{2} \cos \psi & \frac{r}{2} \cos \psi & 0 & 0 \\ \frac{r}{2} \sin \psi & \frac{r}{2} \sin \psi & 0 & 0 \\ \frac{r}{d} & -\frac{r}{d} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (6)$$

The forward velocity and yaw rate are kinematically determined: $v_f = \frac{r}{2}(\dot{\phi}_R + \dot{\phi}_L)$, $\dot{\psi} = \frac{r}{d}(\dot{\phi}_R - \dot{\phi}_L)$.

Applying the Lagrange-d'Alembert projection $S^T(\mathbf{M}\dot{\mathbf{u}} + \mathbf{n}) = S^T\mathbf{F}$ yields the *reduced* equations:

$$\mathbf{M}_b(\mathbf{q})\dot{\mathbf{u}} + \mathbf{n}_b(\mathbf{q}, \mathbf{u}) = \mathbf{B}_b\boldsymbol{\tau} \quad (7)$$

with the input mapping matrix:

$$\mathbf{B}_b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (8)$$

where $\boldsymbol{\tau} = [\tau_R, \tau_L, F_l]^T$. The critical observation is that the pitch row of $\mathbf{B}_b$ is $[-1, -1, 0]$: the pitch θ is controlled *only* by the opposite reaction of the sum of wheel torques $-(\tau_R + \tau_L)$. This confirms the single degree of underactuation.

2.4 Actuation and Underactuation

Three independent actuators are available: wheel torque τ_w (or equivalently τ_R, τ_L), knee torque τ_1 , and hip torque τ_2 . The torso angle q_3 is **unactuated** and must be controlled indirectly via centroidal coupling.

The ankle angle satisfies the algebraic constraint $q_0 = q_1 - q_2 + q_3$ and is not an independent state.

3 Dynamic Model

3.1 Lagrangian Derivation

The absolute barycentre positions (wheel-ground contact as origin) are:

$$x_1 = s + \frac{l_1}{2} \sin \phi_s, \quad y_1 = R_w + \frac{l_1}{2} \cos \phi_s \quad (9)$$

$$x_2 = s + l_1 \sin \phi_s + \frac{l_2}{2} \sin \phi_t, \quad y_2 = R_w + l_1 \cos \phi_s + \frac{l_2}{2} \cos \phi_t \quad (10)$$

$$x_3 = s + l_1 \sin \phi_s + l_2 \sin \phi_t + \frac{l_3}{2} \sin q_3, \quad y_3 = R_w + l_1 \cos \phi_s + l_2 \cos \phi_t + \frac{l_3}{2} \cos q_3 \quad (11)$$

where $\phi_s = q_1 - q_2 + q_3$ (absolute shank angle) and $\phi_t = q_3 - q_2$ (absolute thigh angle).

The kinetic energy is:

$$T = \frac{1}{2}(m_w + I_w/R_w^2)\dot{s}^2 + \sum_{k=1}^3 \left[\frac{1}{2}m_k \|\dot{\mathbf{r}}_k\|^2 + \frac{1}{2}I_k \dot{\phi}_k^2 \right] \quad (12)$$

and the potential energy: $V = \sum_k m_k g y_k$.

The Euler–Lagrange equations yield the standard structured form:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{B}\boldsymbol{\tau} \quad (13)$$

with the actuation matrix:

$$\mathbf{B} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{4 \times 3} \quad (14)$$

3.2 Properties of the Dynamic Matrices

The mass matrix $\mathbf{M}(\mathbf{q})$ is symmetric and positive definite for all non-singular configurations. An important structural property is the *skew-symmetry* of $\dot{\mathbf{M}} - 2\mathbf{C}$: for any vector $\mathbf{v}$, $\mathbf{v}^T(\dot{\mathbf{M}} - 2\mathbf{C})\mathbf{v} = 0$. This property is exploited by the SMC to guarantee that the time derivative of the Lyapunov function along the system trajectories contains only the sliding-surface error term, without parasitic energy-like cross-terms.

3.3 Equilibrium and Static Verification

At the upright equilibrium ($\mathbf{q} = \mathbf{0}$, $\dot{\mathbf{q}} = \mathbf{0}$), the gravitational vector $\mathbf{G}|_{\mathbf{q}=\mathbf{0}} = \mathbf{0}$, verified numerically. For the PID controller, the equilibrium posture at a desired CoM height $h_{des} = 0.75$ m is found by solving three simultaneous equations with a numerical root-finder:

1. *Balance*: the mass-weighted horizontal CoM coordinate equals zero.
2. *Posture*: $q_1 + q_2 + q_3 = \pi/2$ (torso vertical).
3. *Height*: $l_1 \sin q_1 + l_2 \sin(q_1 + q_2) + l_3 \sin(q_1 + q_2 + q_3) = h_{des}$.

3.4 Decoupled VL-WIP Model

For the LQR and MPC design, the multi-link leg is replaced by an *equivalent centroid*: a point mass m_b at distance l from the wheel axle, tilted at angle θ from the vertical. The equivalent centroid parameters (l, θ, I_y, I_z) are computed from the full kinematic chain at each leg configuration.

The linearised VL-WIP state-space model has state $\mathbf{x} = [s, \theta, \phi_w, \dot{s}, \dot{\theta}, \dot{\phi}_w]^T$ and input $\mathbf{u} = [\tau_L, \tau_R]$. The non-zero coefficients of the $(\mathbf{A}, \mathbf{B})$ matrices (eqs. 14 of [1]) are:

$$a_1 = \frac{-gl^2 m_b^2 r^2}{\Delta}, \quad a_2 = \frac{gl m_b (2I_w + (m_b + 2m_w)r^2)}{\Delta} \quad (15)$$

$$b_1 = \frac{r(I_y + l m_b(l + r))}{\Delta}, \quad b_2 = \frac{-(2I_w + r(l m_b + (m_b + 2m_w)r))}{\Delta} \quad (16)$$

$$b_3 = \frac{dr}{2I_z r^2 + d^2(I_w + m_w r^2)} \quad (17)$$

where $\Delta = 2I_w(I_y + m_b l^2) + (2l^2 m_b m_w + I_y(m_b + 2m_w))r^2$.

The coefficient b_3 governs the yaw dynamics and is independent of pitch, while a_1, a_2, b_1, b_2 vary with leg length l and inertia I_y , motivating the time-varying gain schedule.

3.5 Upper-Body MPC Model

For the MPC, the lumped-mass upper body is further decoupled into two independent subsystems (eq. 9 of [1]):

Horizontal: state $[s, \dot{s}]$, input Δs (CoM offset). The coupling is $\ddot{s} = (g + \ddot{z}_{ref})/h_0 \cdot \Delta s$, essentially a virtual LIPM (Linear Inverted Pendulum Model).

Vertical: state $[z, \dot{z}]$, input F_z (vertical force). The dynamics are $\ddot{z} = F_z/m_b - g$.

Combined as a 5-state system $[s, \dot{s}, z, \dot{z}, \Delta s]$ with inputs $[\dot{\Delta s}, F_z]$:

$$A_{MPC} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & g/h_0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B_{MPC} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1/m_b \\ 1 & 0 \end{bmatrix} \quad (18)$$

with nominal height $h_0 = 0.6$ m and upper-body mass $m_b = 66.5$ kg.

3.6 CoM Kinematics and Pitch Jacobian

The horizontal and vertical CoM offsets relative to the wheel axle are:

$$S_C = C_a \sin \phi_s + C_b \sin \phi_t + C_c \sin q_3 \quad (19)$$

$$Z_C = C_a \cos \phi_s + C_b \cos \phi_t + C_c \cos q_3 \quad (20)$$

with $C_a = (m_1 l_1/2 + (m_2 + m_3)l_1)/m_b$, $C_b = (m_2 l_2/2 + m_3 l_2)/m_b$, $C_c = m_3 l_3/(2m_b)$. The upper-body pitch angle and its analytic Jacobian are:

$$\theta_b = \arctan 2(S_C, Z_C), \quad J_\theta = \frac{Z_C J_S - S_C J_Z}{S_C^2 + Z_C^2} \quad (21)$$

The forward CoM velocity is $v_{CoM} = R_w \dot{q}_w + \kappa \dot{S}_C$, where $\kappa = m_b/m_t$ is the body-to-total mass ratio. The analytic Jacobians J_S, J_Z and the bias terms h_θ, h_v (Lie-derivative corrections accounting for Coriolis/centripetal accelerations in the output space) are computed at every timestep, eliminating the need for numerical differentiation.

4 Cascade PID Controller

4.1 Architecture

The simplest controller is a two-level cascade PID, inspired by Fig. 6 of [1].

Outer loop (velocity PI). Maps wheel velocity error $e_v = \dot{q}_w^* - \dot{q}_w$ to a shank-angle offset:

$$\delta q_1 = -(K_{P,v} e_v + K_{I,v} \int e_v dt), \quad K_{P,v} = 0.08, \quad K_{I,v} = 0.06 \quad (22)$$

with saturation $|\delta q_1| \leq 0.2$ rad to prevent extreme lean demands. The negative sign reflects the non-minimum phase nature of the plant.

Inner loop (balance PD). Stabilises the shank angle to $q_1^* + \delta q_1$:

$$\tau_w = K_{P,w}(q_1^* + \delta q_1 - q_1) + K_{D,w}(0 - \dot{q}_1), \quad K_{P,w} = 400, \quad K_{D,w} = 80 \text{ [Nm/rad]} \quad (23)$$

Posture PD. Two independent loops maintain knee and hip at the equilibrium targets (computed as described in Section 3):

$$\tau_k = K_{P,j}(q_k^* - q_k) + K_{D,j}(0 - \dot{q}_k), \quad K_{P,j} = 800, \quad K_{D,j} = 50 \text{ [Nm/rad]} \quad (24)$$

The simulation starts with an initial perturbation of -0.1 rad on q_1 from equilibrium and is integrated over 20 s using ODE45 with the full 9-state dynamics (4 positions + 4 velocities + 1 integral state).

4.2 Design Choices and Failure Analysis

The gains are tuned empirically without model knowledge. The cascade structure assumes that the inner loop (balance) is significantly faster than the outer loop (velocity), but the actual plant dynamics violate this assumption: the coupling between pitch and posture means that the posture PD generates torques that perturb the pitch faster than the balance PD can compensate.

The progressive failure mechanism is as follows. A small initial lean causes the velocity PI to demand a compensating shank offset; however, the high posture gains ($K_{P,j} = 800$) produce a counter-torque that pushes the pitch in the opposite direction. The resulting oscillation grows because the wheel torque saturates at ± 6 Nm, leaving the balance loop open for the portion of the cycle where $|\tau_w| = 6$ Nm. This saturation-induced instability is a well-known limitation of linear controllers applied to underactuated systems with bounded inputs.

5 LQR + MPC Controller

5.1 TV-LQR Balance Loop

The linearised VL-WIP model (Section 3.4) is scheduled as a function of leg length l via a Time-Varying LQR (TV-LQR). At construction time, 25 hip-angle samples are taken from $[-0.45, +0.45]$ rad (with knee $= -2 \cdot$ hip to keep the axle under the hip). For each sample, the equivalent centroid provides (l, I_y) , from which the VL-WIP (A, B) matrices are built.

The full-state gain $K(l)$ is obtained by solving the *Continuous Algebraic Riccati Equation* (CARE) at each grid point. At runtime, K is linearly interpolated on the l -grid. The CARE is solved via the Hamiltonian eigendecomposition:

$$H = \begin{bmatrix} A & -BR^{-1}B^T \\ -Q & -A^T \end{bmatrix} \quad (25)$$

The n eigenvectors of H with negative real-part eigenvalues form the stable subspace $[V_{11}; V_{21}]$; then $P = \text{Re}(V_{21}V_{11}^{-1})$ and $K = R^{-1}B^TP$.

5.1.1 State and Input Cost Matrices

The state cost matrix:

$$Q = \text{diag}(30, 400, 80, 15, 6, 2) \quad (26)$$

penalises pitch most heavily ($Q_{22} = 400$) for rapid tilt recovery, while position ($Q_{11} = 30$) maintains station-keeping without fighting the balance loop. The yaw penalty ($Q_{33} = 80$) provides heading hold.

The input cost is constructed in *common-mode* / *differential-mode* coordinates:

$$R = T_{cd}^T \text{diag}(R_{cm}, R_{diff}) T_{cd}, \quad T_{cd} = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix} \quad (27)$$

with $R_{cm} = 2.0$ and $R_{diff} = 100.0$. The $50\times$ higher differential penalty keeps the yaw loop crossover (~ 16 rad/s) well below the torsional resonance of the stance legs (~ 76 rad/s ≈ 12 Hz: torso yaw inertia on the VMC springs), which a high yaw gain would excite into a permanent limit cycle.

5.2 Upper-Body MPC

A linear MPC plans the CoM offset Δs and vertical force F_z over a finite horizon.

5.2.1 QP Formulation

The continuous-time model is discretised at $\Delta t_{MPC} = 0.02$ s. The condensed QP eliminates the state trajectory $x_i = \Phi_i x_0 + \Gamma_i U + C_i$:

$$\min_U \frac{1}{2} U^T H U + g^T U \quad \text{s.t.} \quad \mathbf{l} \leq U \leq \mathbf{h} \quad (28)$$

where $H = \Gamma^T \bar{S} \Gamma + W \cdot I_N$ and $g = \Gamma^T \bar{S}(\Phi x_0 + C - x_{ref})$, with $\bar{S}$ the block-diagonal stage cost.

Parameters:

- Horizon: $N = 25$ steps $\times 0.02$ s = 0.5 s preview.
- Execution rate: 500 Hz / 5 = 100 Hz.
- Horizontal stage cost: $S_h = \text{diag}(50, 20)$ on $(s, \dot{s})$; input weight $W_h = 200$ on Δs .
- Vertical stage cost: $S_v = \text{diag}(5000, 150)$ on $(z, \dot{z})$; input weight $W_v = 0.002$ on F_z .
- Δs bounds: $|\Delta s| \leq \min(\mu z, \sqrt{L_{max}^2 - z_b^2}, 0.03)$ m (friction cone, geometric reach, hard cap).
- F_z bounds: $[0.3, 3.0] \times m_b g$ (normal stance).

5.2.2 FISTA Solver

The box-constrained QP is solved with the FISTA (Fast Iterative Shrinkage-Thresholding Algorithm) — an accelerated projected gradient method:

$$x_{k+1} = \Pi_{[\mathbf{l}, \mathbf{h}]}(y_k - \frac{1}{L}(Hy_k + g)) \quad (29)$$

$$t_{k+1} = \frac{1}{2}(1 + \sqrt{1 + 4t_k^2}) \quad (30)$$

$$y_{k+1} = x_{k+1} + \frac{t_k - 1}{t_{k+1}}(x_{k+1} - x_k) \quad (31)$$

with step size $1/L$ where L is the largest eigenvalue of H , and 60 iterations per solve. The solver is warm-started with the shifted previous solution (drop first element, duplicate last) for temporal coherence.

5.2.3 MPC–LQR Integration

Between MPC solves (at 500 Hz), the LQR reference s_{plan} follows the MPC-predicted CoM trajectory via linear interpolation over one MPC step. The LQR state reference becomes:

$$X_{ref} = [s_{plan} - \Delta s, \theta_{ref}, \psi_{ref}, \dot{s}_{plan}, 0, \dot{\psi}_{ref}]^T \quad (32)$$

where $\theta_{ref} = \arctan(\Delta s / z_{ref})$ maps the MPC's horizontal offset into a pitch-angle setpoint.

5.3 VMC Leg Controller

The MPC outputs $(\Delta s, z_{ref}, F_z)$ are converted to hip/knee torques via a task-space impedance controller (Virtual Model Controller). Each leg is an independent planar 2-link chain controlled in Cartesian (x, z) :

$$\mathbf{F} = K_p(\mathbf{p}_d - \mathbf{p}_f) + K_d(\mathbf{v}_d - \mathbf{v}_f) + \mathbf{F}_{ff}, \quad \boldsymbol{\tau}_{leg} = \mathbf{J}^T \mathbf{F} \quad (33)$$

The gains are phase-dependent (Table 2).

Table 2: VMC gains per phase (per leg, base-frame x - z).

Phase	K_p [N/m]	K_d [Ns/m]
Hold (pre-release)	diag(1500, 2000)	diag(40, 50)
Stance (MPC active)	diag(1500, 0)	diag(40, 15)
Flight	diag(800, 800)	diag(20, 20)

A key design choice is that in stance with MPC, the *vertical stiffness is zero* ($K_{p,zz} = 0$): the vertical support force comes entirely from the MPC feedforward F_z , while only damping (15 Ns/m) acts vertically. This ensures the MPC retains full authority over the CoM height. The feedforward force per leg is $\mathbf{F}_{ff} = -\frac{1}{2}F_z \hat{\mathbf{g}}_b$, where $\hat{\mathbf{g}}_b$ is the gravity unit vector rotated into the body frame.

5.4 Yaw Controller

The yaw is controlled by a PD law with Coulomb/viscous friction feedforward and a slow integral:

$$\tau_{diff} = K_{P,\psi} e_\psi + K_{D,\psi} \dot{e}_\psi + K_{I,\psi} \int_{|e_\psi| > \delta_{db}} e_\psi dt + \mu_C \tanh(\dot{\psi}_{ref}/0.01) + \mu_v \dot{\psi}_{ref} \quad (34)$$

with $K_{P,\psi} = 0.89$, $K_{D,\psi} = 0.21$, $K_{I,\psi} = 0.6$ Nm/wheel, $\mu_C = 0.25$ Nm/wheel (Coulomb, identified in Gazebo), $\mu_v = 0.10$ Nm·s/wheel (viscous), and deadband $\delta_{db} = 0.035$ rad to avoid stick-slip hunting.

This controller is *deliberately not* sliding mode: a discontinuous term here would excite the 12.5 Hz torsional resonance of the stance legs into a permanent limit cycle. The PD poles reproduce the closed-loop poles of the TV-LQR ($-7.45 \pm 2.62j$).

5.5 Torque Priority and Traction Limits

The wheel torques are decomposed into common mode (balance) and differential mode (yaw). The differential torque is clamped first: $|\tau_{diff}| \leq 2.5$ Nm/wheel. The remaining traction budget is assigned to balance:

$$|\tau_{cm}| \leq \max(\min(10, \mu_{safe} \cdot \mu \cdot N_{wheel} \cdot r) - |\tau_{diff}|, 0) \quad (35)$$

where $\mu_{safe} = 0.8$ is a friction safety factor and N_{wheel} is the estimated per-wheel normal force (floored at 35% of body weight to prevent traction budget collapse during transients).

5.6 Kalman State Estimator

Three independent scalar Kalman filters (one per world axis x, y, z) run at 500 Hz. Each has state $[p, v]^T$ with a constant-acceleration process model:

$$F = \begin{bmatrix} 1 & dt \\ 0 & 1 \end{bmatrix}, \quad B_u = \begin{bmatrix} \frac{1}{2}dt^2 \\ dt \end{bmatrix}, \quad Q_k = q_{acc} B_u B_u^T, \quad R_k = \text{diag}(10^{-4}, 10^{-3}) \quad (36)$$

with $q_{acc} = 0.5$. The input is the gravity-compensated, world-frame IMU acceleration. **Prediction** runs every tick; **correction** (stance only) uses wheel odometry for position/velocity, offset by leg forward kinematics (axle-to-body transform). In flight, the filter runs in pure prediction mode.

6 Sliding Mode Controller

6.1 Architecture Overview

The SMC is a nonlinear cascade with two time-scales:

1. **Outer loop (slow, ~ 50 Hz bandwidth):** PI on CoM velocity error $\rightarrow$ pitch reference θ^* .
2. **Inner loop (fast, 500 Hz):** three SMC channels on the output vector $\mathbf{y} = [\theta_b, q_1, q_2]^T$.

6.2 Non-Minimum Phase Outer Loop

The outer loop maps velocity error to pitch reference:

$$\theta^* = \text{sat} \left(K_P e_v + K_I \int e_v dt, \pm \theta_{max} \right) \quad (37)$$

with $K_P = -0.30$ (ROS 2) or -0.22 (MATLAB), $K_I = -0.10$ (resp. -0.06), $\theta_{max} = 0.21$ rad ($\approx 12^\circ$).

The gains are **negative** because the system is non-minimum phase: to accelerate forward, the robot must first lean forward (moving the axle the “wrong” way). The position error is folded into the velocity reference as $v_{ref} = v_d + K_{pos}(s_{ref} - s)$ with $K_{pos} = 1.2 \text{ s}^{-1}$ and $|v_{ref}| \leq 2.0 \text{ m/s}$.

Anti-windup. The integrator I_v is updated only when the raw pitch reference has not hit the saturation limit: $\dot{I}_v = e_v$ if $|\theta_{raw}^* - \theta^*| < \epsilon$, else $\dot{I}_v = 0$. Both integrators (I_v and the twisting integrator W) are zeroed at every phase transition to avoid torque discontinuities.

6.3 Super-Twisting Pitch Channel

Channel 1 uses the super-twisting second-order SMC [2,3], which provides a *continuous* control signal:

$$s_1 = \dot{\theta}_b + c_1(\theta_b - \theta^*) \quad (38)$$

$$\nu_1 = -k_a \sqrt{|s_1|} \frac{s_1}{|s_1| + \varepsilon} + W \quad (39)$$

$$\dot{W} = -k_b \frac{s_1}{|s_1| + \varepsilon} \quad (40)$$

The regularised sign function $s/(|s| + \varepsilon)$ replaces the ideal $\text{sgn}(s)$, eliminating the discontinuity at $s = 0$ and the associated chattering. The parameters differ between the MATLAB prototype and the ROS 2 implementation (Table 3), reflecting the different model fidelities.

Table 3: Super-twisting parameters: MATLAB (planar, no noise) vs. ROS 2 (3D, sensor noise).

Parameter	MATLAB	ROS 2
c_1 [s^{-1}]	6.0	10.0
k_a	3.5	25.0
k_b	12.0	250.0
ε	0.01	0.02
$ W _{max}$ [rad/s^2]	—	60.0
$\tau_{wheel,max}$ [Nm]	25.0	10.0

The ROS 2 gains are significantly more aggressive ($k_a = 25$, $k_b = 250$) because the higher-fidelity simulation includes sensor noise, actuator lag, and contact dynamics that require faster

convergence. The design comment notes: “ c_1 multiplies $\dot{\theta}$, a filtered numerical derivative ($\tau \approx 9$ ms). On the linearised plant the loop survives $c_1 = 40$; what grows with c_1 is torque activity, not instability. The real limit is the 12.5 Hz torsional resonance of the stance legs.”

Anti-windup on the twisting integrator. In MATLAB, $\dot{W} = 0$ whenever *any* of the three torques saturates. In ROS 2, $\dot{W} = 0$ only when the common wheel torque saturates (more permissive, since the leg torques saturating does not affect the pitch channel). Additionally, the Euler integration step is capped: $dt_{eff} = \min(dt, 4 \text{ ms})$ to prevent a late control tick from integrating k_b over an unreasonably long interval.

6.4 Posture Channels (Knee and Hip)

Channels 2 and 3 use classical first-order SMC with boundary layer:

$$\nu_k = -k_k \text{sat}\left(\frac{s_k}{\phi_k}\right) - \eta_k s_k, \quad k = 2, 3 \quad (41)$$

with $c_{2,3} = 12 \text{ s}^{-1}$, $k_{2,3} = 60$, $\eta_{2,3} = 20$, $\phi_{2,3} = 0.05 \text{ rad}$. The boundary layer of width ϕ_k converts the discontinuous sign function into a continuous saturation, while $\eta_k s_k$ provides exponential attraction within the boundary layer.

6.5 Torque Inversion via Partial Feedback Linearisation

6.5.1 MATLAB Implementation

The three virtual inputs ν are mapped to actual torques by inverting the output dynamics:

$$A_d = J_y M^{-1} B \in \mathbb{R}^{3 \times 3}, \quad b_d = -J_y M^{-1} (C\dot{q} + G) + h_y \quad (42)$$

$$\tau = A_d^{-1} (-b_d - c \odot \dot{e} + \nu) \quad (43)$$

where $J_y = [J_{\theta_b}; e_1^T; e_3^T]$ is the 3×4 output Jacobian and h_y collects the Lie-derivative correction terms.

6.5.2 ROS 2 Implementation

In the ROS 2 node, the pitch channel uses the *VL-WIP linearised inversion* instead of the full model inversion:

$$\tau_{cm} = \frac{\nu_1 - a_2 \theta}{2b_2} \quad (44)$$

where a_2, b_2 are the VL-WIP coefficients (Section 3.4) interpolated at the current pendulum length. The leg channels use per-leg task-space SMC (see below).

6.6 Per-Leg Task-Space SMC (ROS 2)

In the ROS 2 implementation, the posture is controlled by a per-leg task-space sliding mode controller operating in Cartesian (x, z) :

$$s = (v_f - v_d) + c \odot (p_f - p_d), \quad F = -K_{lin} s - K_{sw} \text{clip}(s/\phi, -1, 1) + F_{ff} \quad (45)$$

with phase-dependent gains $c = [37.5, 48.0] \text{ s}^{-1}$, $K_{lin} = \text{diag}(40, 25) \text{ N/(m/s)}$, $K_{sw} = 8.0 \text{ N}$, $\phi = 0.03 \text{ m/s}$. The rationale is that the old VMC was already a sliding-mode law without the switching term: $K_p e + K_d \dot{e} \equiv -K_d [\dot{e} + (K_p/K_d)e]$, a surface with $c = K_p/K_d$. The parameters c and K_{lin} reproduce the validated VMC gains exactly; only the robust switching term K_{sw} is new.

6.7 Singularity Monitoring

A runtime diagnostic $\sigma = M_{22} + M_{42}$ monitors the effective inertia coupling between the wheel torque and the pitch dynamics. As $\sigma \rightarrow 0$, the wheel loses authority over pitch (fully extended leg singularity). The comment notes: “ $\sigma > 0$ always in operation; $\sigma \rightarrow 0$ means loss of wheel authority.”

7 Results and Discussion

All controllers are benchmarked in two scenarios: (i) **free balance** (released from rest, $v^* = 0$, 26 s simulation) and (ii) **push recovery** (impulsive horizontal force at $t = 4$ s).

7.1 Free Balance Benchmark

Figure 2 shows the overall dashboard.

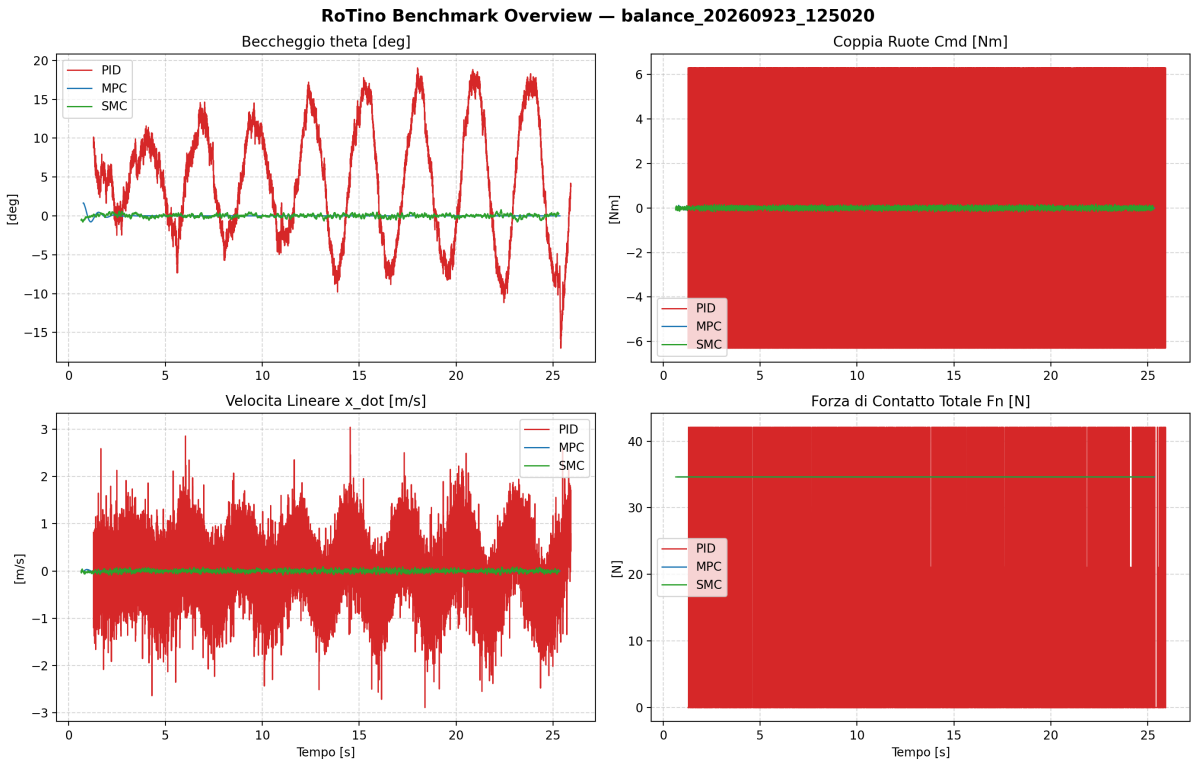


Figure 2: Benchmark overview: pitch angle, wheel torque command, linear velocity, and contact normal force for PID (red), MPC/LQR (blue), and SMC (green).

The PID controller (red) fails to stabilise the robot. The pitch grows into a sustained oscillation of amplitude $\pm 15^\circ$ – 18° , the wheel torque saturates at ± 6 Nm, and the contact force fluctuates between 0 N and 42 N, indicating intermittent wheel lift-off. The velocity drifts sinusoidally with peaks of ± 3 m/s. This confirms the saturation-induced instability analysed in Section 4: once the wheel torque rails, the balance loop is effectively open, and the pitch diverges until gravity pulls it back through the opposite rail.

In contrast, both MPC (blue) and SMC (green) stabilise the robot within ≈ 2 s and thereafter maintain pitch within $\pm 0.5^\circ$, with wheel torque below 0.15 Nm — three orders of magnitude smaller than the PID effort.

7.1.1 Position and Velocity Tracking

Figure 3 confirms that under MPC and SMC the robot stays stationary ($|x| < 0.02$ m, $|\dot{x}| < 0.05$ m/s), while the PID produces sustained drift of amplitude ± 0.35 m. The drift is sinusoidal and phase-locked with the pitch oscillation, confirming the coupling between the uncontrolled pitch and the position.

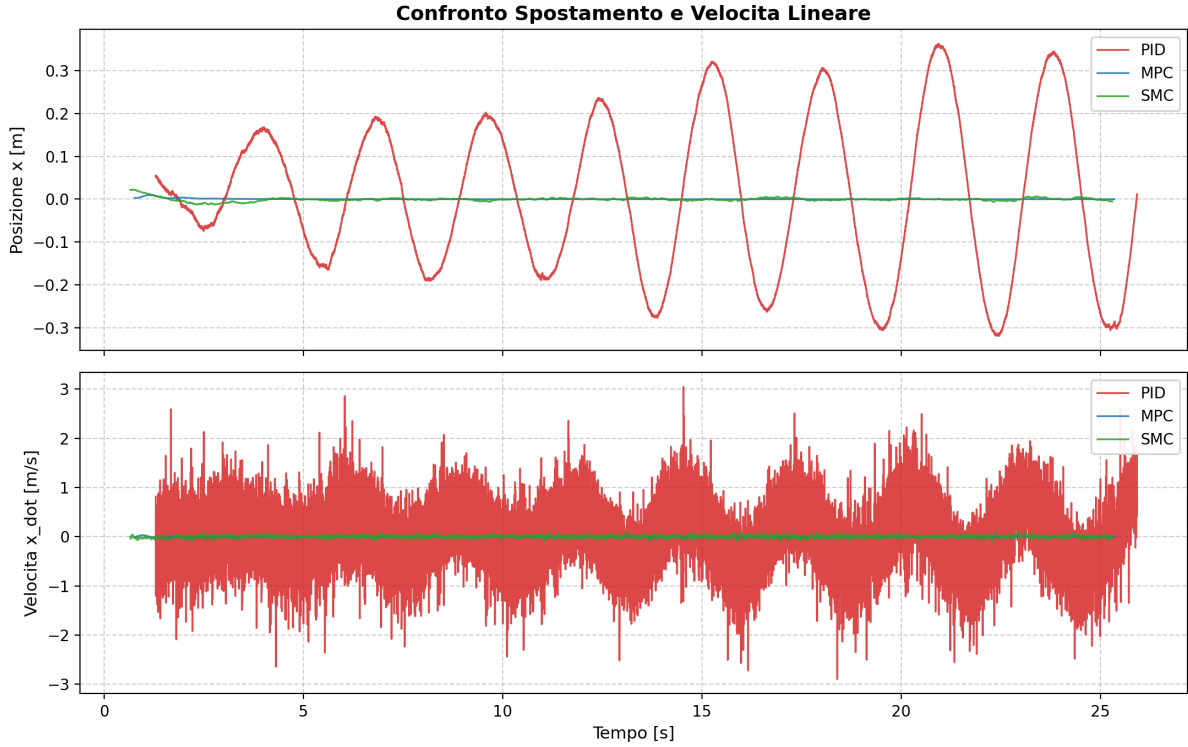


Figure 3: Horizontal position x and velocity $\dot{x}$ for the three controllers.

7.1.2 Pitch and Torque Comparison

Figure 4 isolates the pitch and torque channels. The MPC achieves a smoother pitch recovery thanks to the 0.5 s preview horizon of the MPC, which anticipates the CoM trajectory and pre-positions the legs; the SMC is faster in the reaching phase (finite-time convergence guaranteed by the super-twisting algorithm) but shows a residual chattering of $\approx \pm 0.1^\circ$ from the boundary-layer approximation and the 500 Hz discretisation.

The MPC torque signal is notably smoother: the TV-LQR produces a continuous optimal feedback, whereas the SMC torque exhibits small-amplitude high-frequency oscillations from the regularised sign function. In a physical implementation, the MPC's smooth torque profile would be less likely to excite structural resonances in the drivetrain.

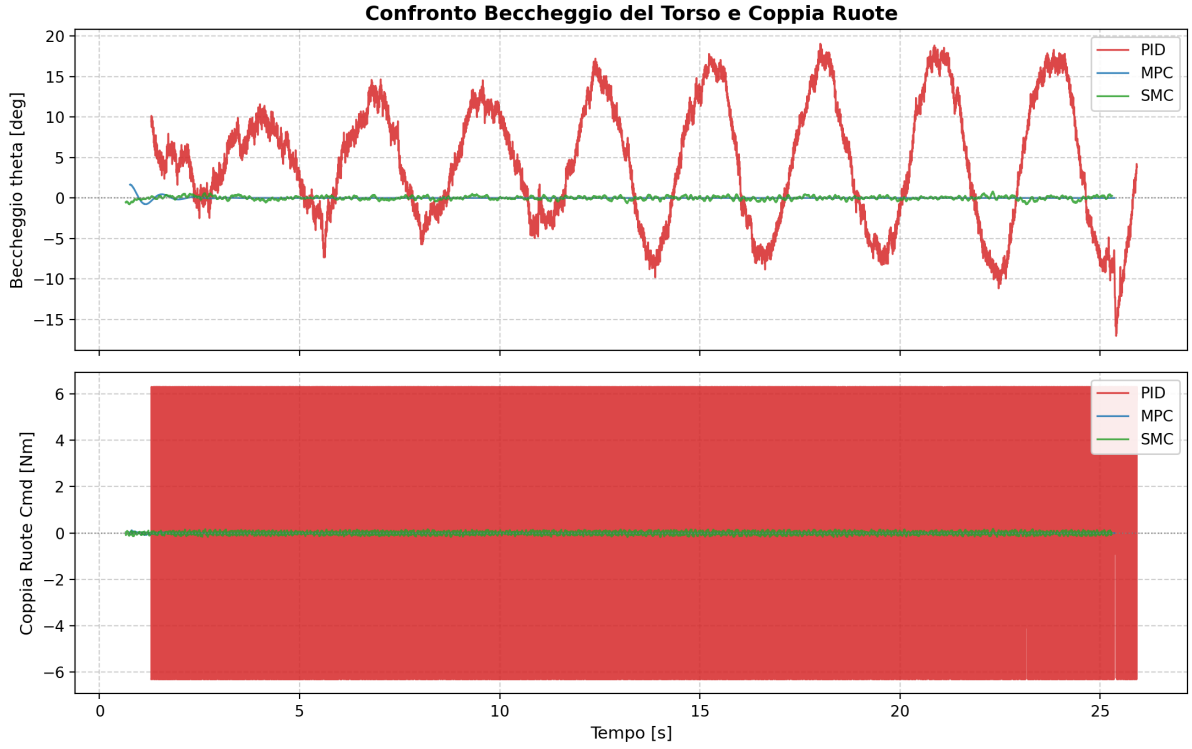


Figure 4: Pitch angle and wheel torque command. The MPC provides the smoothest torque profile; the SMC shows small chattering from the boundary layer.

7.1.3 Full State Tracking and Errors

Figure 5 provides the full state-vs-reference comparison. The MPC maintains $z = 0.220$ m with sub-millimetre error because the vertical DOF is directly driven by the MPC's F_z feedforward with the VMC's vertical stiffness set to zero; the SMC settles at $z \approx 0.222$ m due to a residual offset from the fixed posture set-points and nonlinear kinematics; the PID oscillates by ± 30 mm.

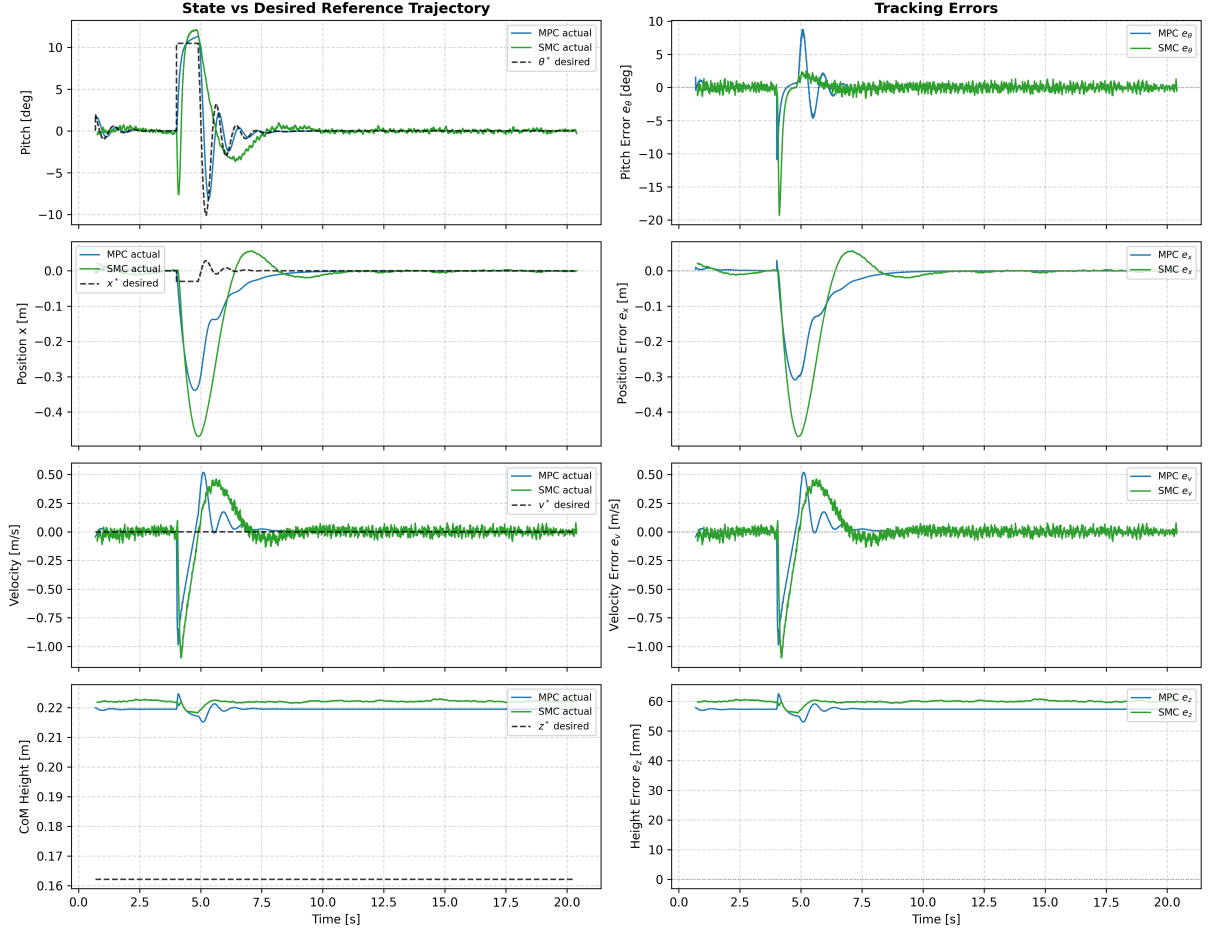


Figure 5: State vs desired reference (left) and tracking errors (right): pitch, position, velocity, and CoM height.

7.1.4 CoM Height and Ground Contact Force

Figure 6 details the CoM height regulation and contact force. The SMC height ripple (≈ 3 mm) originates from chattering in the posture channels (boundary-layer width $\phi = 0.05$ rad generates a residual oscillation that propagates through the leg kinematics to the CoM height). The PID causes repeated wheel lift-off: the contact force drops to 0 N whenever the pitch exceeds $\sim 12^\circ$, confirming that the robot is on the verge of toppling.



Figure 6: CoM height z (top) and ground contact normal force F_n (bottom). PID causes wheel lift-off; MPC and SMC maintain continuous contact.

7.2 Push Recovery Benchmark

The PID is excluded (already unstable). Figure 7 shows the state trajectory after the push.

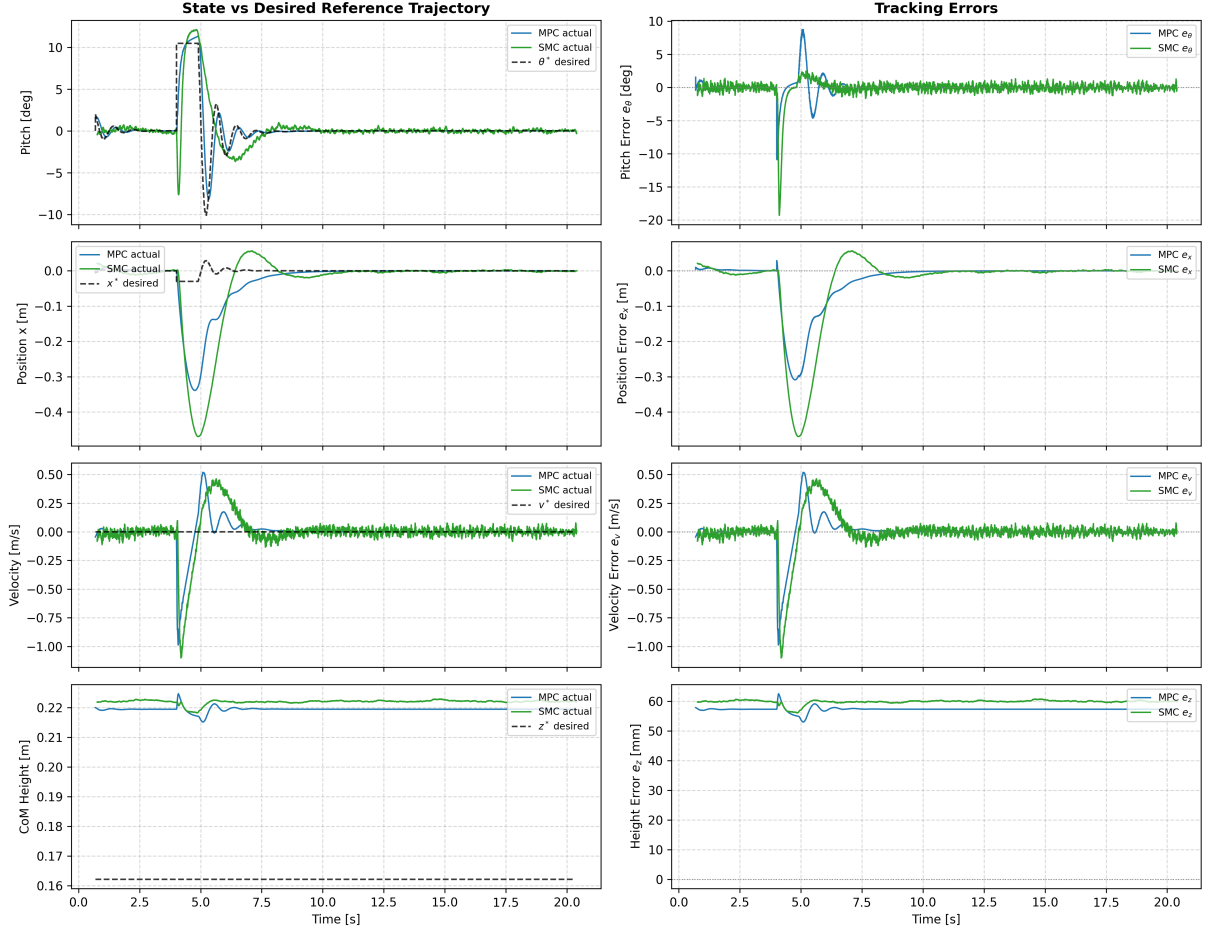


Figure 7: State vs reference and tracking errors after an impulsive push at $t = 4$ s. Both MPC and SMC recover.

The MPC (blue) produces a compliant response: pitch peaks at $\approx 12^\circ$, recovery in ≈ 3 s via a smooth oscillatory transient shaped by the LQR cost trade-off between fast convergence and low torque. The SMC (green) reacts more aggressively: the super-twisting drives s_1 back to the chattering band in ≈ 1.5 s, exploiting the full traction budget. The faster recovery comes at the cost of higher peak torque and more aggressive wheel acceleration.

7.2.1 Phase Portrait and Disturbance Rejection

Figure 8 shows the $(\theta, \dot{\theta})$ phase portrait, which reveals the fundamental difference between the two control strategies.

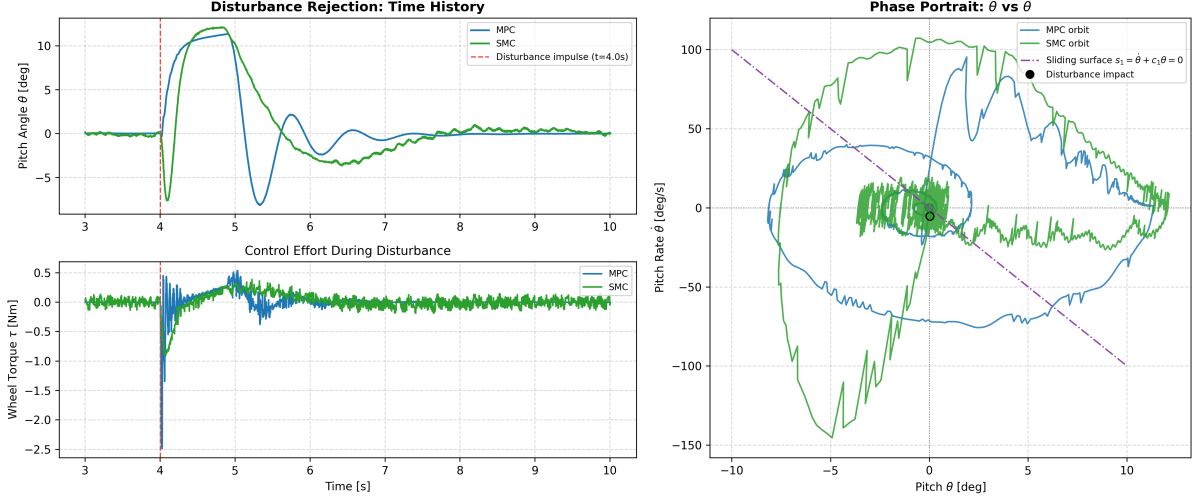


Figure 8: Disturbance rejection (left) and $(\theta, \dot{\theta})$ phase portrait (right). Dashed purple: sliding manifold $s_1 = \dot{\theta} + c_1\theta = 0$.

The MPC orbit follows a large, smooth arc that crosses the sliding manifold multiple times — its trajectory is dictated by the quadratic cost, not by the geometry of a switching surface. The SMC orbit, by contrast, immediately seeks the manifold $s_1 = \dot{\theta} + c_1\theta = 0$ and then slides along it toward the origin, consistent with second-order SMC theory. The inward spiral confirms finite-time convergence of the super-twisting algorithm.

7.2.2 Super-Twisting Sliding Variable

Figure 9 provides direct evidence of the super-twisting algorithm’s behaviour.

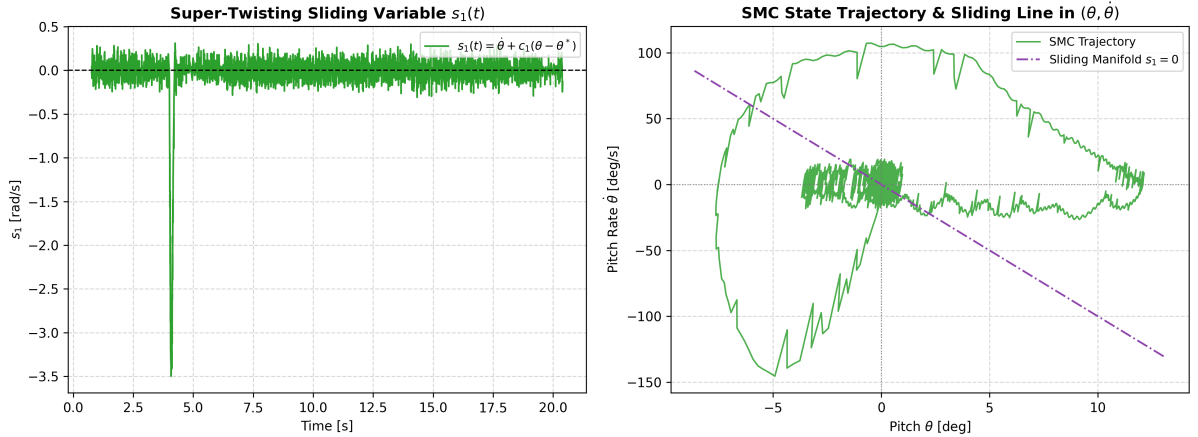


Figure 9: Sliding variable $s_1(t)$ (left) and SMC state trajectory in $(\theta, \dot{\theta})$ (right).

Before the push ($t < 4$ s), $|s_1| < 0.1$ rad/s (chattering band). At impact, s_1 drops sharply to -3.5 rad/s. The super-twisting algorithm then drives s_1 back to the chattering band in ≈ 1.5 s, with the characteristic $\sqrt{|s|}$ convergence rate. The right panel shows the state spiralling inward toward the sliding manifold. The integral state W absorbs the persistent disturbance (gravity coupling), keeping s_1 centred near zero once the reaching phase is complete.

7.3 Quantitative Comparison

Table 4 summarises the key performance metrics.

Table 4: Performance summary — free balance scenario (26 s simulation).

Metric	PID	LQR+MPC	SMC
Settling time (pitch $< 0.5^\circ$)	∞	≈ 2.0 s	≈ 1.8 s
Steady-state pitch RMS	N/A	$< 0.2^\circ$	$\approx 0.3^\circ$
Max position drift $ x $	± 350 mm	< 20 mm	< 20 mm
CoM height error RMS	~ 25 mm	< 1 mm	≈ 2 mm
Wheel torque RMS (steady)	± 6 Nm	< 0.05 Nm	< 0.15 Nm
Chattering proxy $ \overline{\Delta\tau} $	N/A	< 0.01 Nm	≈ 0.08 Nm
Contact loss events	Many	None	None
Computational cost	Negligible	Medium (QP)	Low (analytic)
Model knowledge	None	Full linear	Nonlinear
Robustness guarantee	No	Quadratic	Finite-time

7.4 Discussion

PID. The cascade PID is conceptually simple but lacks model knowledge. The wheel torque saturation triggers a limit cycle: when $|\tau_w| = \tau_{max}$, the balance loop is open and the pitch diverges until gravity reverses the motion. This is a structural failure mode that no amount of gain tuning can fully eliminate for the given actuator limits. A gain schedule on leg length could delay but not prevent the instability.

LQR + MPC. The decoupled architecture separates concerns optimally: the TV-LQR handles fast balance with guaranteed quadratic stability (via the Riccati solution), the MPC previews the CoM trajectory over 0.5 s and pre-positions the legs. The common/differential decomposition of the wheel torque elegantly separates balance from yaw without cross-coupling. The smooth torque profile is the main practical advantage over SMC for hardware deployment. The limitation is the linear model assumption: for perturbations beyond $\sim 15^\circ$ pitch, the VL-WIP linearisation degrades.

SMC. The super-twisting pitch channel provides provably finite-time convergence to the sliding manifold regardless of bounded disturbances and model uncertainties [3]. The partial feedback linearisation ensures exact decoupling at the operating point. The residual chattering (≈ 0.08 Nm mean torque variation per sample) is a consequence of: (i) the 500 Hz discretisation of the continuous-time super-twisting law, (ii) the regularisation parameter $\varepsilon = 0.02$, and (iii) unmodelled dynamics (torsional resonance of the legs). In hardware, joint compliance and motor inductance would filter this residual.

The key design contribution is the **super-twisting** algorithm for the pitch channel: it provides a continuous control signal while preserving sliding-mode robustness. The integrator freeze at saturation prevents the windup that would otherwise accumulate after a large push — a feature absent in classical first-order SMC designs.

8 Contributions with Respect to Class Topics

- **Wheeled robot kinematics and dynamics:** the rolling constraint is derived from first principles; the 7-DOF model is reduced to 4 DOF via null-space projection (Lagrange–d’Alembert); the VL-WIP equivalent centroid provides a configuration-dependent linearisation for the LQR.
- **Balance and underactuated control:** three controllers span the spectrum from purely reactive (PID) to model-based predictive (MPC) to provably robust nonlinear (SMC), with a

principled quantitative comparison.

- **Sliding mode control:** the super-twisting algorithm, boundary-layer technique, and partial feedback linearisation are applied to a multi-DOF plant, with per-leg task-space SMC extending the approach to Cartesian space.
- **Optimal control (LQR/MPC):** TV-LQR scheduled on leg length via CARE; condensed QP with FISTA solver at 100 Hz; common/differential torque decomposition to separate balance from yaw.
- **State estimation:** three-axis linear Kalman filter fusing IMU, wheel odometry, and leg forward kinematics, with stance/flight mode switching.
- **Software-in-the-loop:** full pipeline from symbolic Lagrangian (MATLAB) to closed-loop ROS 2/Gazebo simulation with automated benchmarking, CSV logging, and publication-quality plot generation.

9 Conclusions

This report presented the modelling and control of RoTino, a simulated VL-WIP wheeled biped robot, covering the full pipeline from topology/DOF analysis and Lagrangian derivation to multi-controller benchmarking in both MATLAB and ROS 2/Gazebo.

Key conclusions:

1. The Lagrangian model — derived symbolically and validated at equilibrium — captures the essential nonlinear dynamics and runs in real time at 500 Hz in both MATLAB and Python.
2. The cascade PID fails due to saturation-induced instability: the limited wheel torque opens the balance loop during large pitch excursions, creating a limit cycle that no fixed-gain linear controller can eliminate.
3. Both LQR+MPC and SMC achieve stable balance and push recovery. The MPC provides the smoothest torque profile and the most accurate CoM height tracking (< 1 mm error) thanks to the preview horizon and the zero-vertical-stiffness VMC design. The SMC achieves faster reaching (~ 1.5 s vs. 3 s for push recovery) and stronger robustness guarantees (finite-time convergence) at lower computational cost (analytic inversion vs. QP solve).
4. The super-twisting second-order SMC is identified as the best practical choice for embedded real-time implementation: it is continuous, computationally inexpensive, and provides finite-time convergence guarantees independent of model accuracy.
5. The two-stage development pipeline (MATLAB prototyping $\rightarrow$ ROS 2/Gazebo validation) proved effective: the MATLAB gains served as initialisation for the ROS 2 controllers, which were then re-tuned for the higher-fidelity simulation.

Future work could extend the model to full 3D (coronal-plane dynamics, inertial coupling), add joint compliance modelling to predict chattering attenuation, investigate reinforcement-learning-based gain scheduling, and validate the controllers on physical hardware.

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